

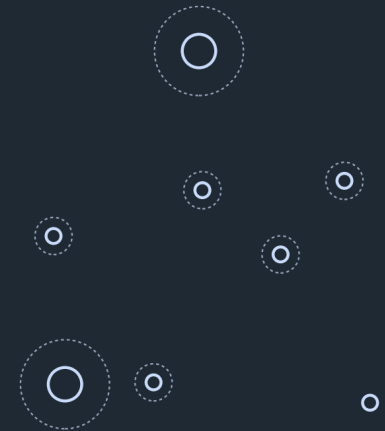
CrossMEP: A Tiered Synthetic Dataset of Multi-Trade MEP Cross-Sections

A support-design problem distribution for learning-based structural support assembly synthesis

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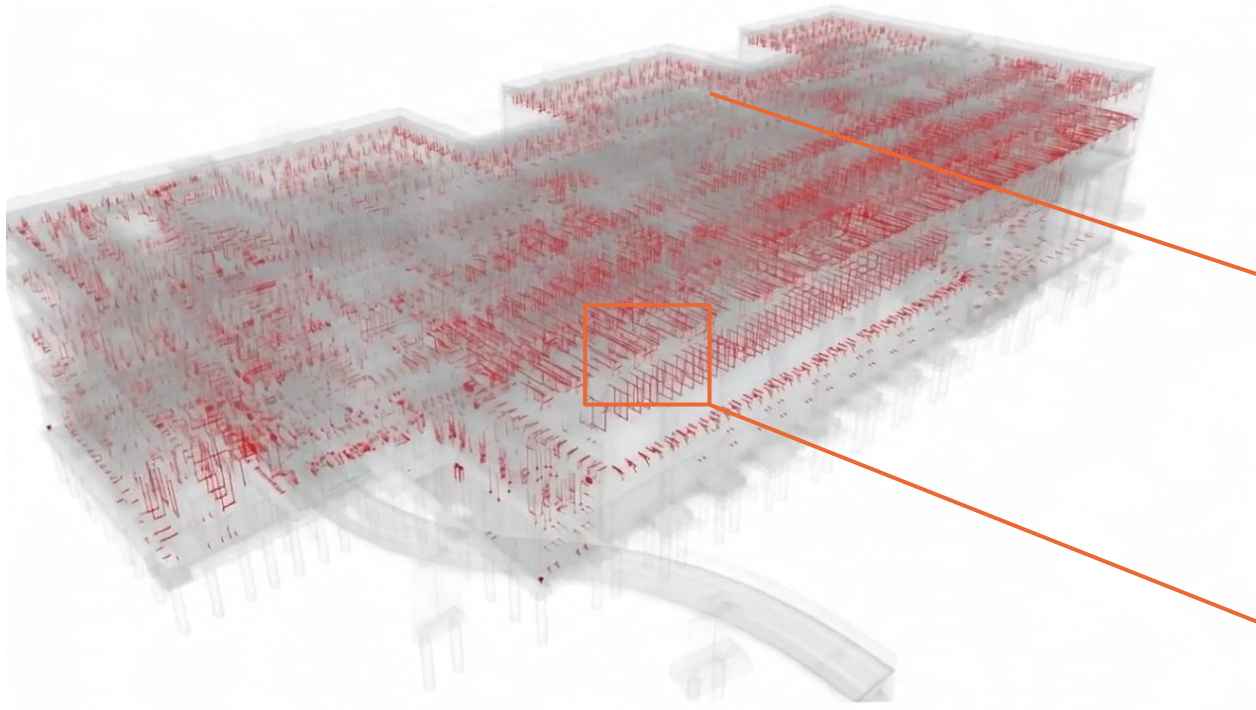
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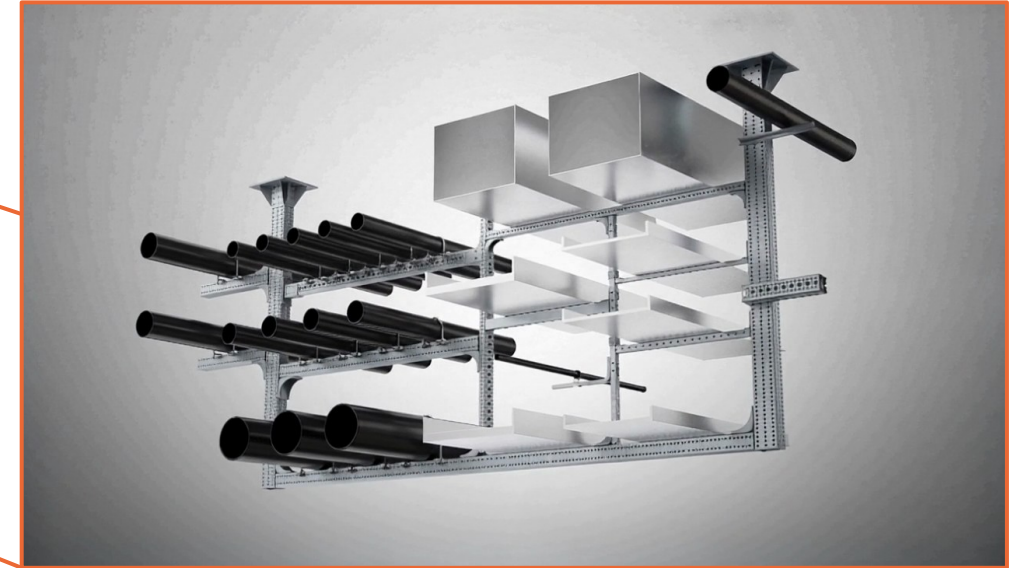


a generated C8 context: eight services on three rows

A hospital needs about 10,000 support assemblies, each designed by hand



1 unit = 1 structural support assembly (SSA)



$\approx 10,000$

support assemblies in one 200,000 sq ft hospital



20 min – 2 h

of engineering for each assembly, designed one by one



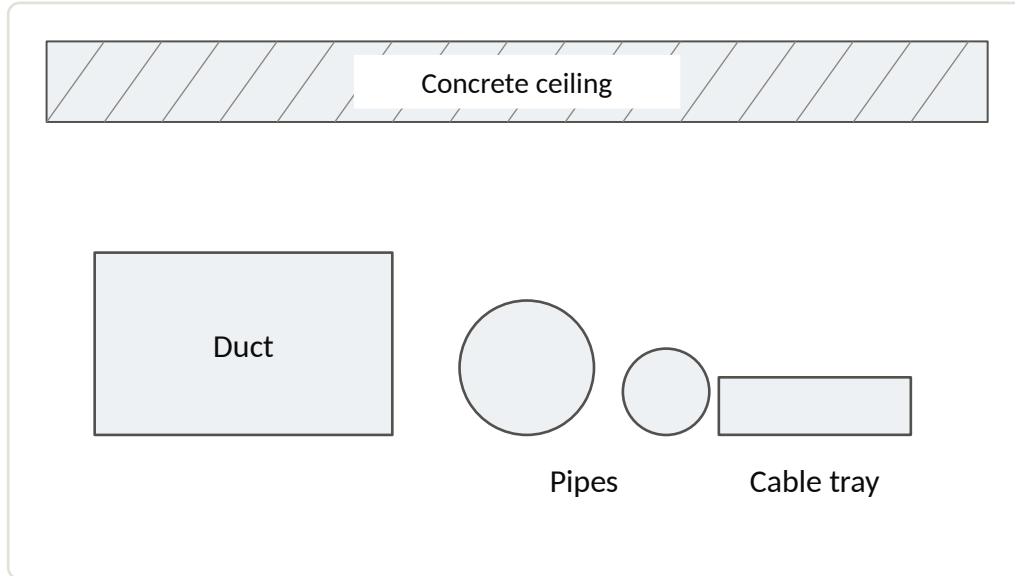
$\approx \frac{1}{4}$

of MEP design effort; on the order of \$600K per project

Approximate practitioner estimates [1]. [1] Pedrollo, Graeber & Fischer (2026), Feasibility-aware sequential synthesis of structural support assemblies, ISARC 2026

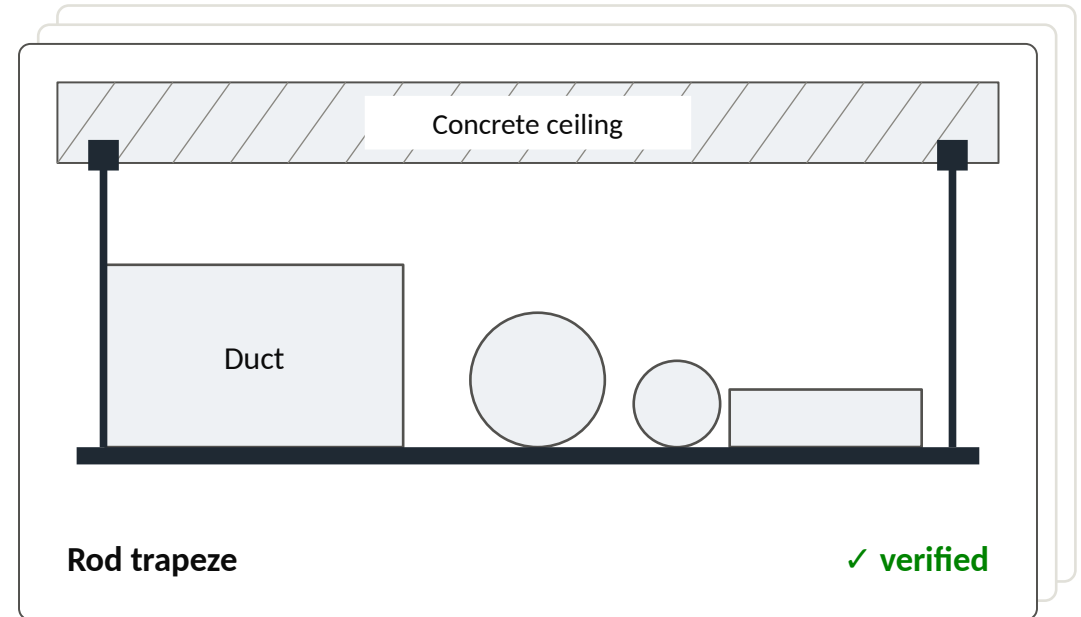
A designer turns one cross-section into verified support assemblies

In: one cross-section = a context



Structure: ceiling, floor or wall, with anchor zones
Services: trade, position, size and weight per metre

Out: verified support designs = assemblies

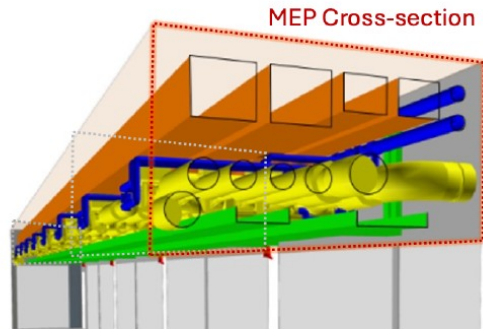


Up to ten designs, ranked by installed cost
Each checked: statics, anchors, connectors, buildability

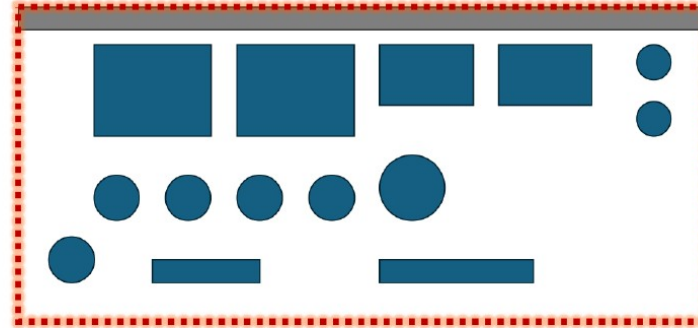
Context, the brief: the section at one hanger, its structure and services.
This is what CrossMEP contains.

Assembly, the answer: catalog parts that carry the services to the structure. Not in CrossMEP: a method produces it.

Synthesizing the assembly from a catalog and a context has resisted automation



(a) Full MEP route



(b) One CrossMEP context

Design starts after coordination [3]: at each hanger the designer works from the section across the run, not from the whole model [2].



Where practice stands

Tools coordinate the model and check a design.

Choosing the topology and the parts, from a fixed catalog, is still done by hand, one section at a time [1].



1. Geometric judgment

Engineers choose an assembly's topology from experience.



2. Rule explosion

Code, load and material constraints interact across thousands of combinations.

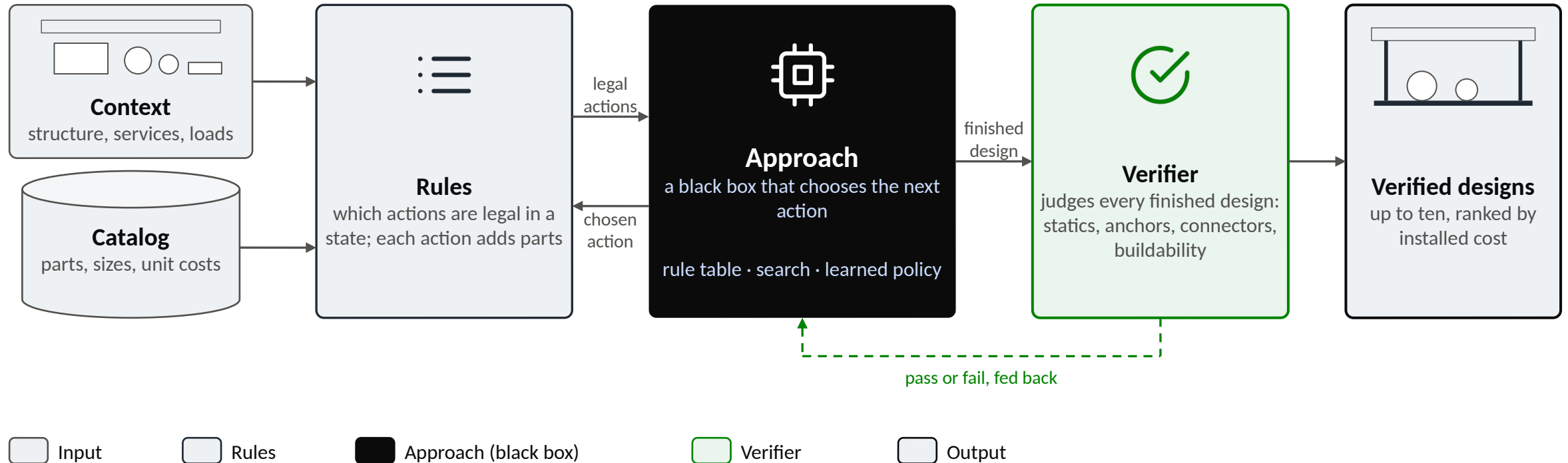


3. Catalog volatility

Catalogs change faster than rule-based systems can be rewritten.

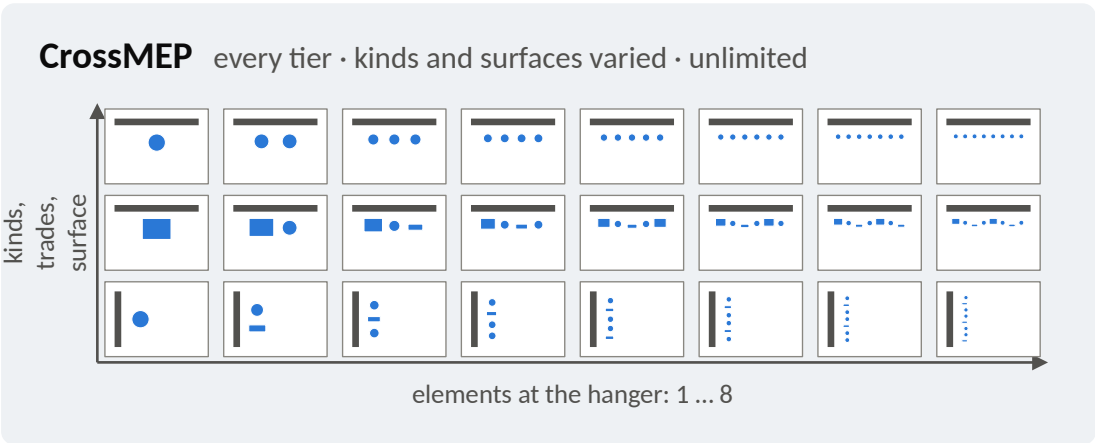
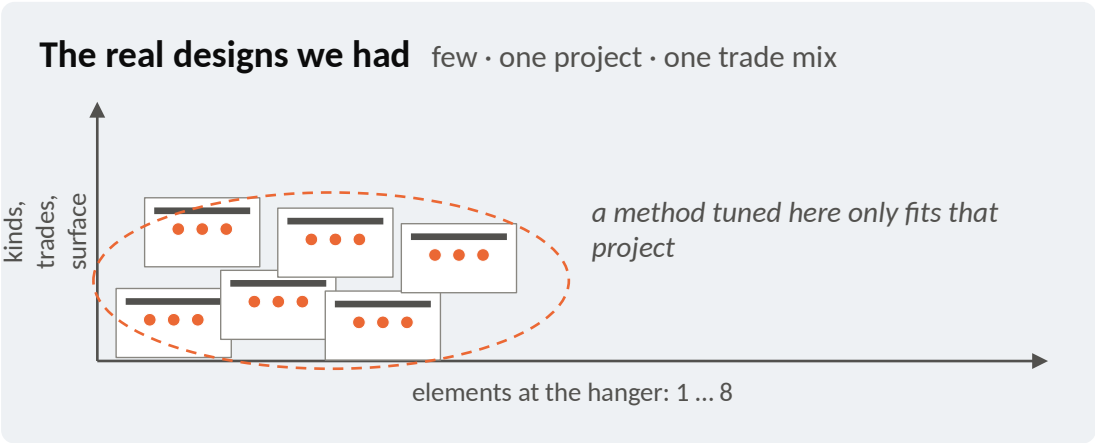
[1] Pedrollo, Graeber & Fischer (2026), Feasibility-aware sequential synthesis of structural support assemblies, ISARC 2026 [2] Pedrollo, Gvadzabia, Graeber & Fischer (2026), CrossMEP, CIB W78 2026 (this paper) [3] Korman, Fischer & Tatum (2003), Knowledge and reasoning for MEP coordination, J. Constr. Eng. Manage. 129(6)

Whatever the method, it needs problems to learn from and a fixed set to compare on

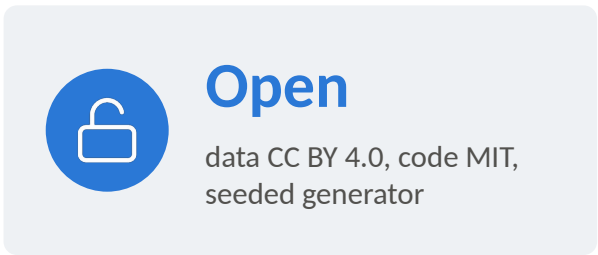
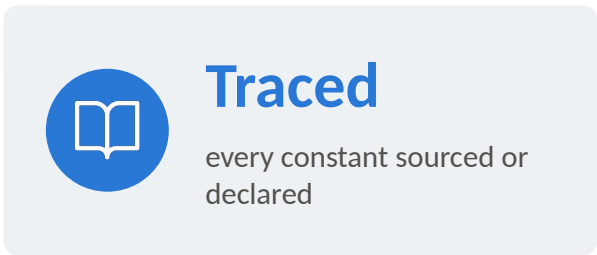
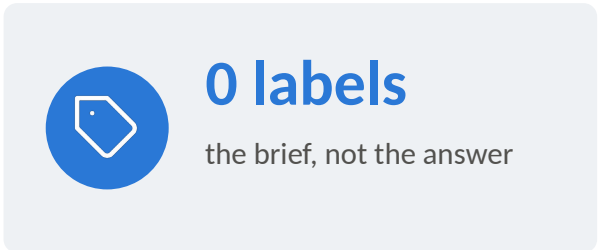
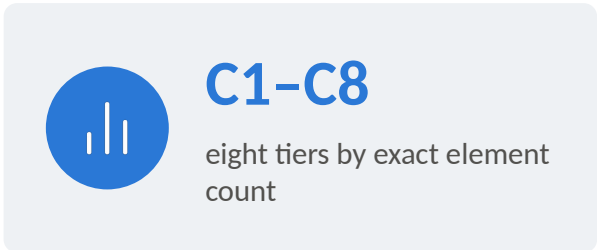
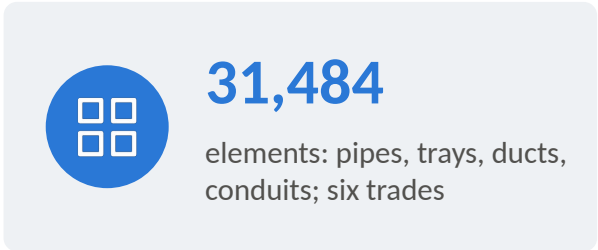


Whatever sits in the black box, it needs **problems to learn from**, a **fixed set to compare on** and **problems that look like practice**.

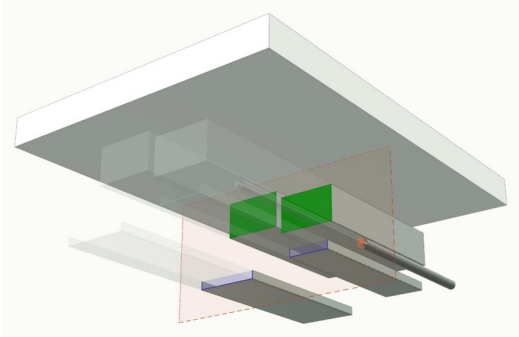
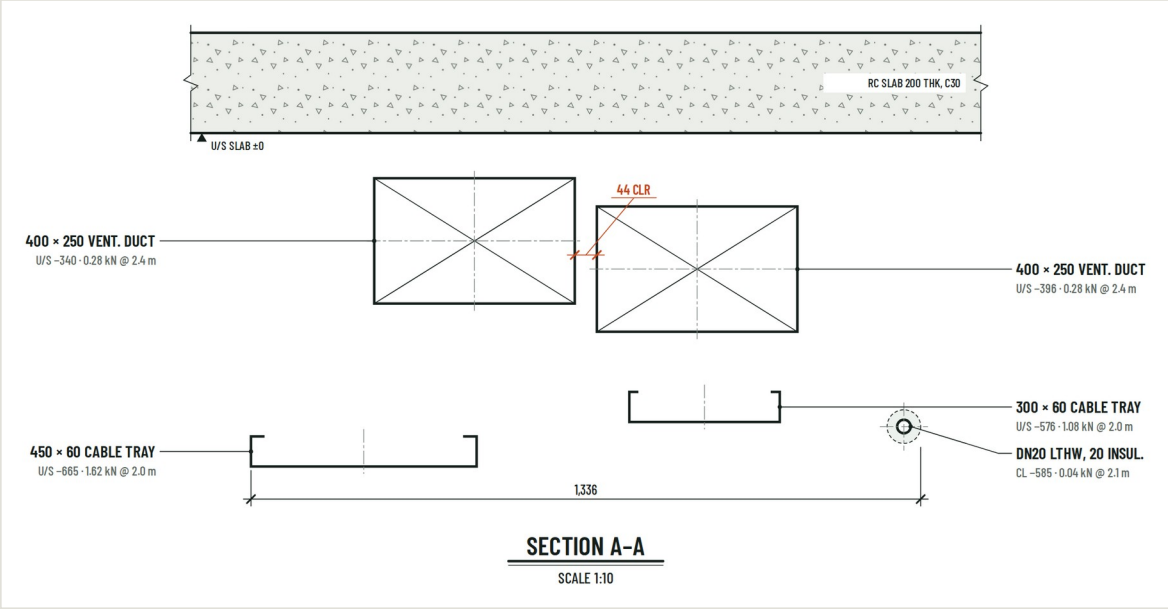
Real designs cover one corner of the design space; CrossMEP covers all of it



CrossMEP at a glance



We generate what a designer receives: the section at one hanger



● ventilation ● electrical ● heating

What every element carries

- kind
- service, trade
- size
- insulation
- position
- load = kN/m × span

What the support designer is given

Element	Trade	Ins. mm	kN/m	Span	kN at support
400×250 duct	● ventilation	0	0.115	2.4 m	0.28
400×250 duct	● ventilation	0	0.115	2.4 m	0.28
450 tray	● electrical	0	0.809	2.0 m	1.62
300 tray	● electrical	0	0.540	2.0 m	1.08
DN20	● heating	20	0.019	2.1 m	0.04



200 mm
ceiling slab,
concrete C30



3.30 kN
total load at this
support



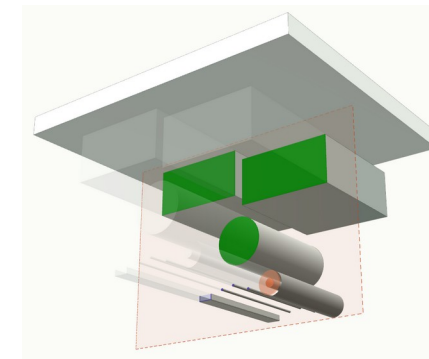
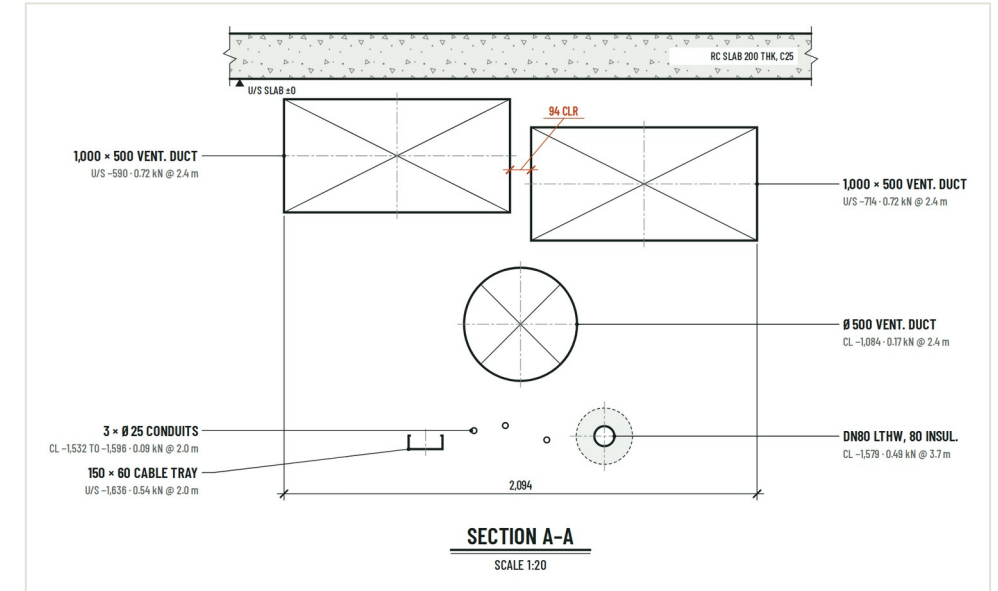
44 mm
closest clear gap,
2 rows



Not given, by design: channel, rods, clamps, anchors. No correct answer either: a feasible support depends on the catalog you build from.

Sections follow trade practice and measured spacing, not random shapes

- 1 **# Tier**
Cn fixes the exact element count, $n = 1 \dots 8$
- 2 **Surface**
ceiling or wall (78 / 22 %); concrete, 150–300 mm thick
- 3 **Services**
grouped as trades run [3]: hot + cold, flow + return, conduit groups, 1–2 trays, ducts
- 4 **Rows**
bulky first: ducts nearest the slab, then trays and conduits, then pipes
- 5 **Spacing**
gaps drawn from a built project [12], at least 25 mm; electrical above wet on walls [10]
- 6 **Check**
every pair 25 mm clear, rows centred, everything recorded; same seed, same file



● ventilation ● electrical ● heating

One generated context, tier C8: 8 services on 3 rows, ducts nearest the slab, then the tray and conduits, then the pipe. The gap marked on the sheet is one draw from the measured distribution; the same seed gives the same section.

[3] Korman, Fischer & Tatum (2003), Knowledge and reasoning for MEP coordination, J. Constr. Eng. Manage. 129(6) [10] IET On-Site Guide (BS 7671), conduit support spacing; BS 7671 Reg. 528.3.2 [12] buildingSMART International (2020), Duplex Apartment and Medical-Dental Clinic sample IFC files, CC BY 4.0

Sizes, spans and loads come from standards; the choices are declared



Pipes DN15–150

Sizes EN 10220 / EN 10255 medium series [4]

Load steel + water × ASME B31.1 water-service span [8]



Insulation

Sizes GEG Anlage 8, heated lines [11]; 30 / 50 mm condensation control, chilled

Load adds to the size, not the load



Cable trays 150–600

Sizes IEC 61537 systems [6] in NEMA VE 1 widths [7]

Load 50 kg/m full at 300 mm × 2.0 m, below the shortest NEMA class span [7]



Ducts

Sizes EN 1505 rectangular, EN 1506 round [5]

Load manufacturer weight table with flanges × 2.4 m (SMACNA) [9]



Conduits Ø20–50

Sizes IEC 61386-1 [6], parallel groups of 2–6

Load steel tube + 40 % cable fill × 2.0 m (IET / BS 7671) [10]



Standards

what each element is and weighs: sizes, walls, spans, insulation [4]–[11]



Measured

how close neighbours sit: clear gaps and row stagger from two open buildings [12]



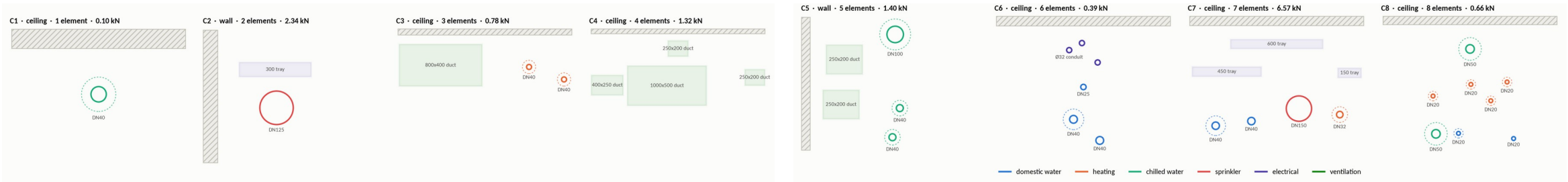
Declared

our choices, labelled as such: trade mix, ceiling / wall 78 / 22 %, number of rows

Every constant carries its source and status in the repository; the verification log says why each standard was chosen over its alternatives.

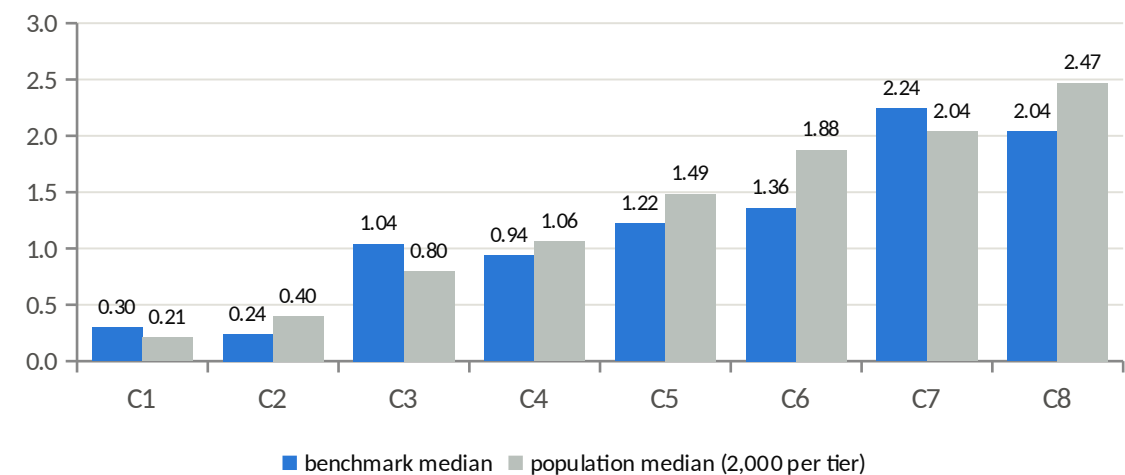
[4] EN 10220:2002, EN 10255:2004 (steel tubes: dimensions, medium series) [5] EN 1505:1997, EN 1506:2007 (sheet-metal air ducts: rectangular and circular dimensions) [6] IEC 61386-1 (conduit systems), IEC 61537 (cable tray and ladder systems) [7] NEMA VE 1 (metal cable tray systems: widths, load-class spans) [8] ASME B31.1 Table 121.5 (hanger spacing), as republished in the ASHRAE Handbook, HVAC Systems and Equipment [9] SMACNA, HVAC Duct Construction Standards, Metal and Flexible, Table 5-1 [10] IET On-Site Guide (BS 7671), conduit support spacing; BS 7671 Reg. 528.3.2 [11] GEG (Gebäudeenergiegesetz), Anlage 8 (pipe insulation thickness) [12] buildingSMART International (2020), Duplex Apartment and Medical-Dental Clinic sample IFC files, CC BY 4.0

Tier Cn holds exactly n elements; higher tiers are heavier and tighter, as designed

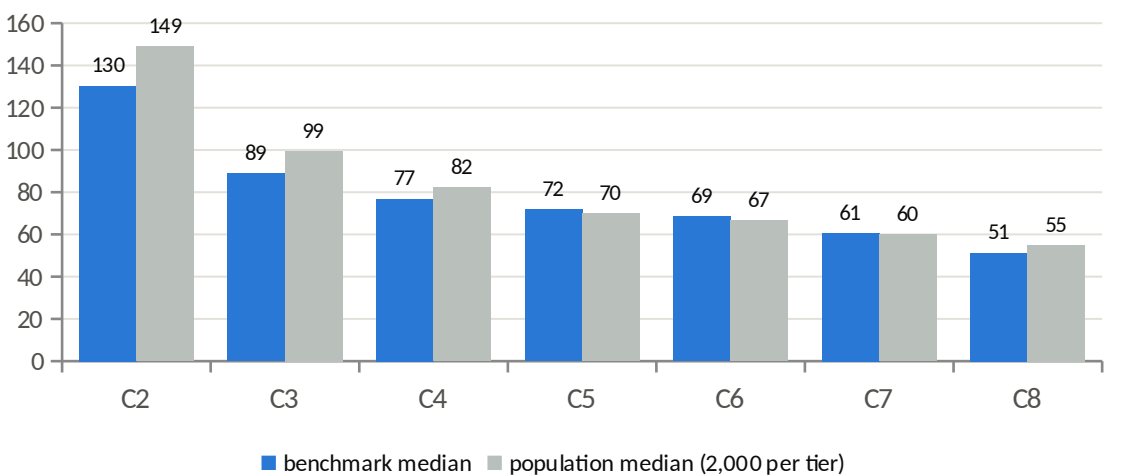


One benchmark context per tier, C1 to C8; dashed rings are insulation. Within a tier, kinds, trades, surfaces and stacking all vary: the element count is the one controlled axis of difficulty.

Heavier with the tier: median load at the support (kN)



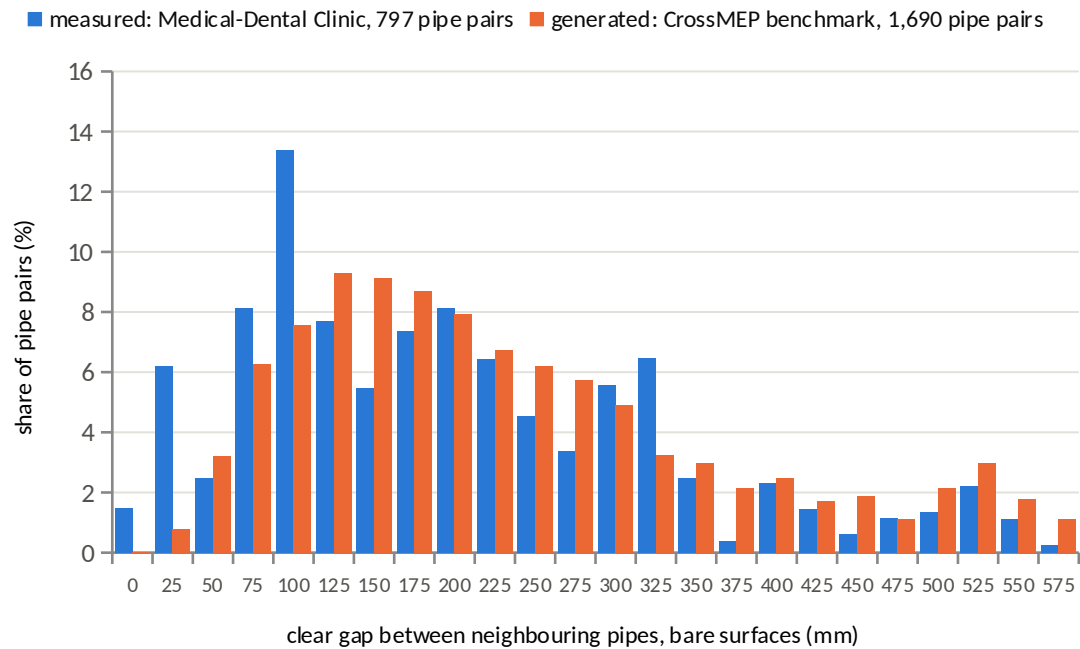
Tighter with the tier: median closest clear gap (mm)



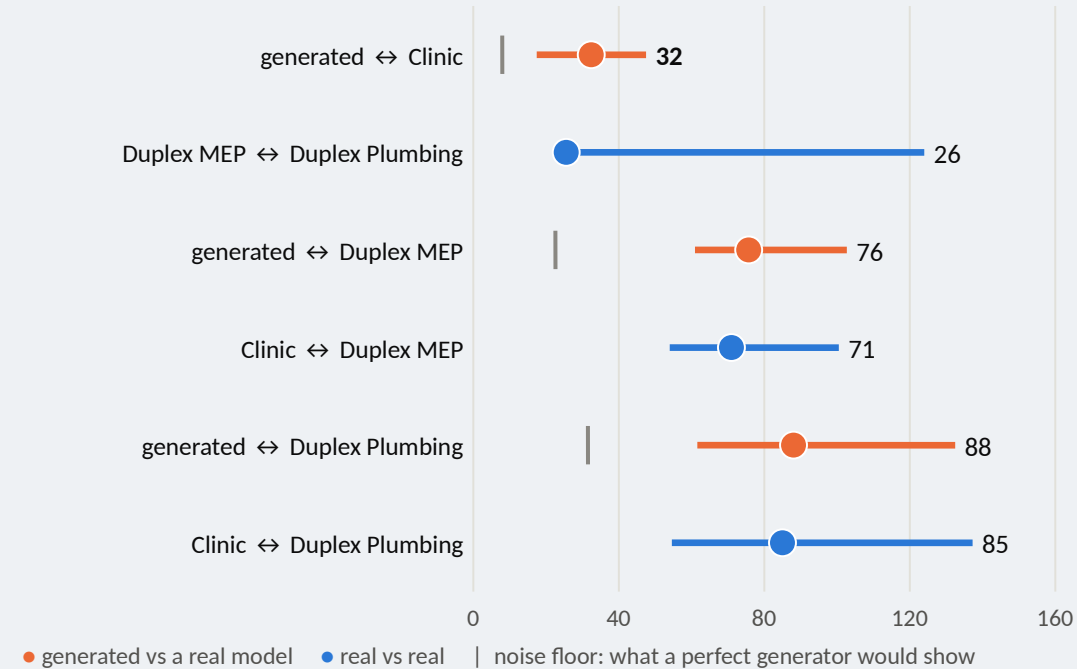
A check that the dataset behaves as designed, not a discovery: the tier fixes the count; load and congestion follow from the rules. Load rises with the tier (rank correlation +0.46 on the benchmark; the population medians rise at every step) and the closest gap narrows at every step (-0.38). Bootstrap intervals in RESULTS.md.

Spacing holds up on a clinic the generator never saw

A building the generator never saw



How far apart the distributions are: Wasserstein-1 (mm) with 95 % interval



32 mm

generator to clinic (95 % CI 17–48); fitted on the duplex, clinic held out

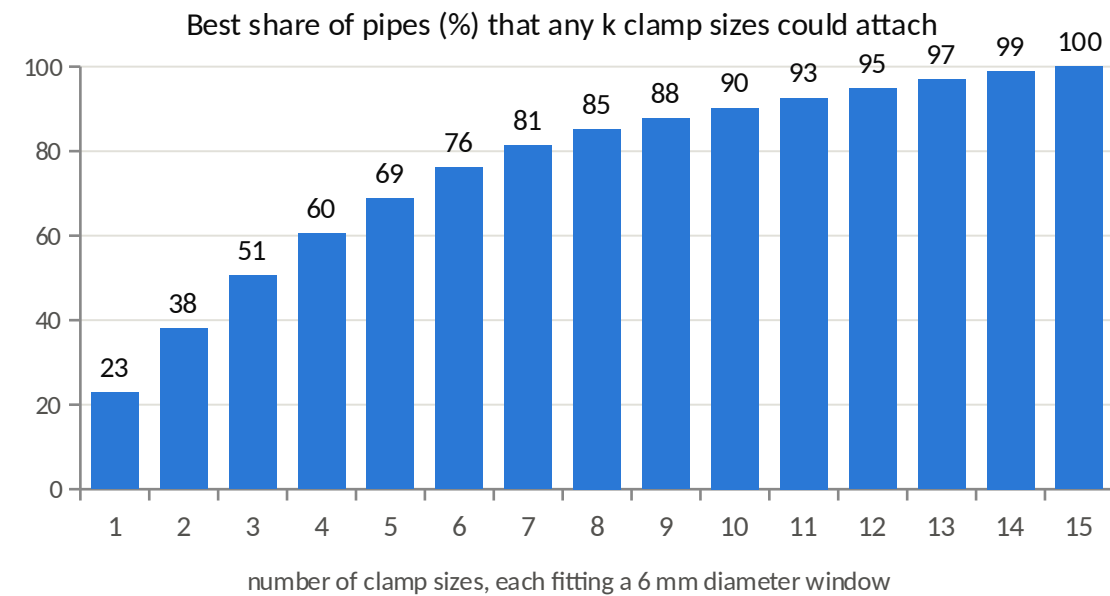
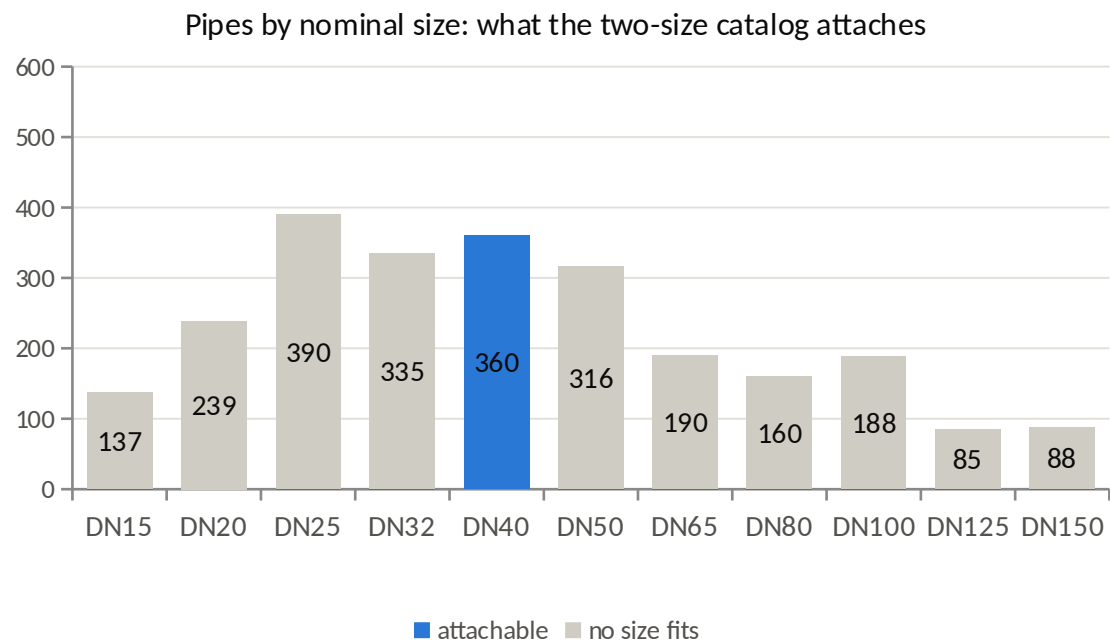
26 mm

the duplex's own two models from each other: about as close as real is to real

190 mm

what a fixed 25 mm gap would be from the clinic

A two-size catalog attaches 1 pipe in 7; the dataset shows what to cover



14.5 %

of benchmark pipes attachable with the paper's two sizes [2] (95 % CI 12.7–16.2); every one is DN40

38.1 %

with two sizes placed where the pipes are; 76 % with six, and every pipe with 15

0 of 2,012

trays, ducts and conduits: a pipe-clamp catalog defines no attachment. Load never binds (1.97 vs 2.5 kN)

[2] Pedrollo, Gvadzabia, Graeber & Fischer (2026), CrossMEP, CIB W78 2026 (this paper)

Set the parameters and the generator returns the designer's brief

SERVICES AT THE HANGER

By tier

Exact mix

TIER = NUMBER OF SERVICES

C1

C2

C3

C4

C5

C6

C7

C8

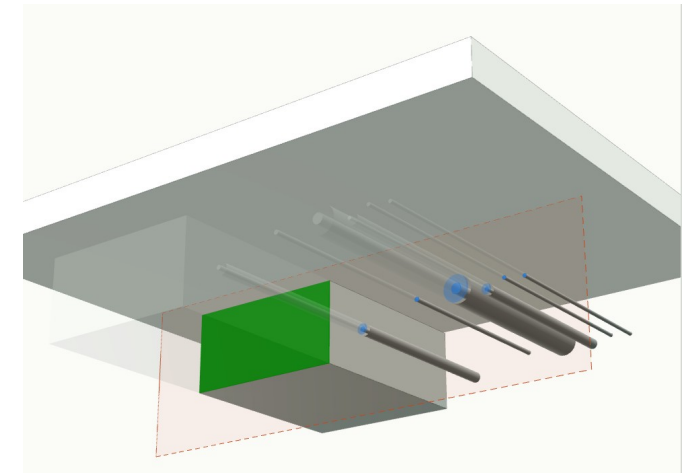
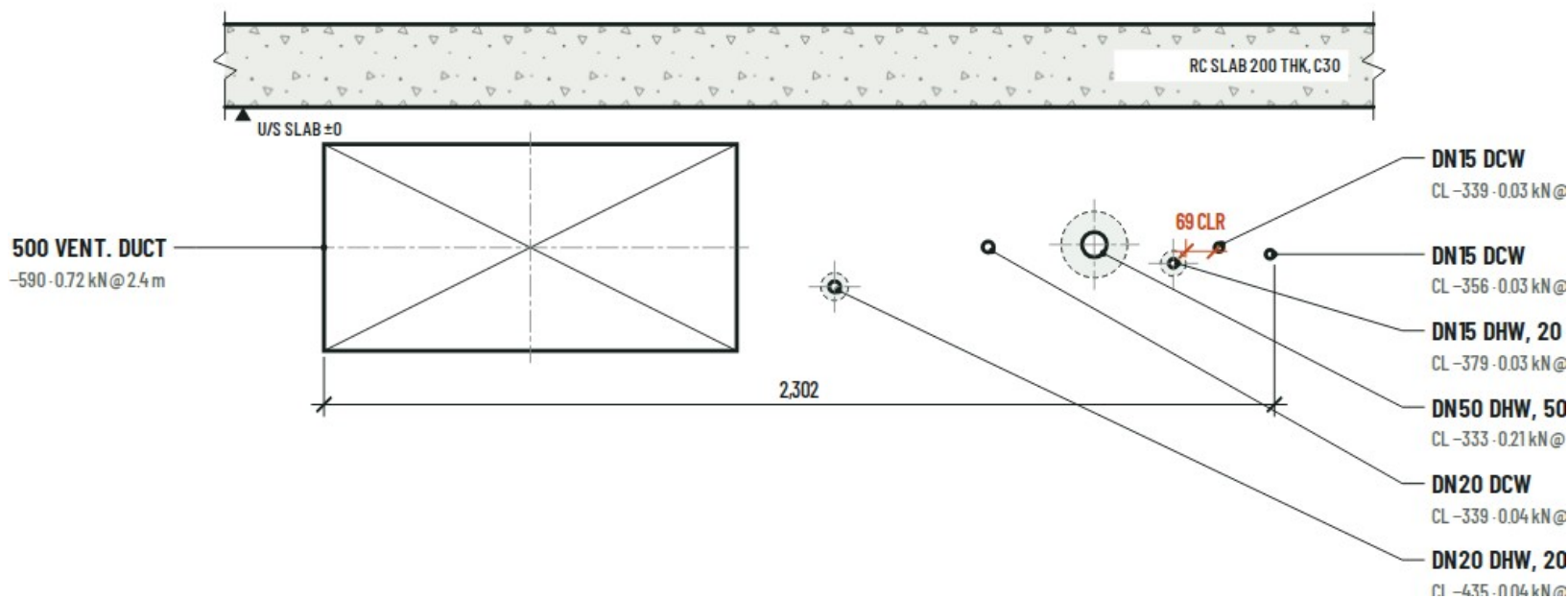
SEED

<

0

>

Generate another



Scan to open the studio

In: the parameters (tier C7, seed 0). Out: section A-A at the support, as the support designer receives it (scale 1:20), and the same run in 3D. Every section is a stored output of the released generator, with the Python call that reproduces it.

Train on the generator, evaluate on the benchmark, report per tier

```
import crossmep.tasks as cm
data = cm.load("benchmark") # 1,000 contexts
cm.filter_contexts(data, pipes=2, trays=1)

from crossmep.evaluate import score, compare
score(my_results, data) # per tier, 95 % CI
compare(mine, baseline, data) # paired test
```

Loading, metrics and scoring need only the Python standard library; the generator needs NumPy. Command line: `python -m crossmep generate | validate | evaluate`.

Four splits on disjoint seeds; the test seeds are never trained on

5,000

train
seed 1000

500

validation
seed 2000

500

test
seed 3000

1,000

benchmark
seed 42 · 125 per
tier

7,000 contexts, 31,484 elements; tiers assigned round-robin. Plain JSON with a JSON Schema, Croissant metadata and a datasheet.

Train

on the generator: unlimited seeds, a curriculum from C1 up to C8

Evaluate

on the benchmark: 125 held-out contexts per tier, the same for every method

Report

per tier with 95 % intervals, paired tests between methods

One section at one support today; a checker and real data come next

Scope

- One cross-section at one support: routing, branches and elevation changes are outside it; so is support spacing — the span is recorded per element, so a method can vary it
- Realism checked on sections of two open buildings, a residential duplex and a medical clinic; congested racks rest on coordination practice and standards
- Trade mix and tier composition are design choices, not survey results
- Synthetic: not for the structural design of real installations

Next

- A public checker for support designs, with best-known costs from an exact solver on the small tiers
- The catalog as an input: generated catalogs, and tests on catalogs a method has never seen
- A real test set: sections from commercial projects with supports designed by engineers

If you coordinate services or design supports, open the gallery, filter for the scenes you know, and tell us what looks wrong. Every rule is one documented constant away from being changed.



Support design now has open problems to learn from

7,000 support-design problems · every constant sourced or declared
pipe gaps checked on two open buildings · open data and code

Data CC BY 4.0, code MIT. Not for the structural design of real installations.



github.com/lavinia-ped/CrossMEP_Dataset
lavinia@stanford.edu